value of V incorrectly as g/ω = 9.81/1.53 = 6.4 m s⁻¹.

3 (a) (i) gravitational force provides centripetal force B1

$GMm / r^2 = mv^2 / r$ B1

$v = \sqrt{\frac{GM}{r}}$ A1

Comments: For a "show" question, should start with a statement. Also the word "provides" needs to be written for full credit.

(ii) gain in kinetic energy = loss in gravitational potential energy B1

$\frac{1}{2} mV_0^2 - 0 = 0 - (-GMm / x)$ C1

$V_0 = \sqrt{\frac{2GM}{x}}$ A1

Comments: When question ask "Explain your working", full spelling is needed, e.g. "gravitational potential energy" rather than just "GPE". Leniency in marking saved the day for many candidates in this prelim but please spell in full in A level for qualitative or "show" questions. Besides that, a lot of candidates did not put negative sign for gravitational potential energy. Again, Leniency in marking saved the day for them.

(iii) for $x = r$, V_0 is greater than v M1

Stone could not enter orbit A1

Comments: A lot of candidates did not realise that when V_0 is greater than *v, the centripetal force required is greater than the gravitational force provided, and hence gravitational force is insufficient to provide for centripetal force, thus no circular motion. Leniency in marking saved the day for many candidates.*

(b) tangent drawn at (4.0, 14.5) B1

gradient = 3.3×10^{-24} to 3.9×10^{-24} A1

field strength = $(3.6 \times 10^{-24}) / (1.6 \times 10^{-19})$

$= 2.3 \times 10^{-5}$ V m^{-1} A1

Comments: Some students gave the force instead of the field strength as the answer.

4 **(a)** **(i)** $T = 2\pi\sqrt{(2.3 / 63)} = 1.20$ s C1